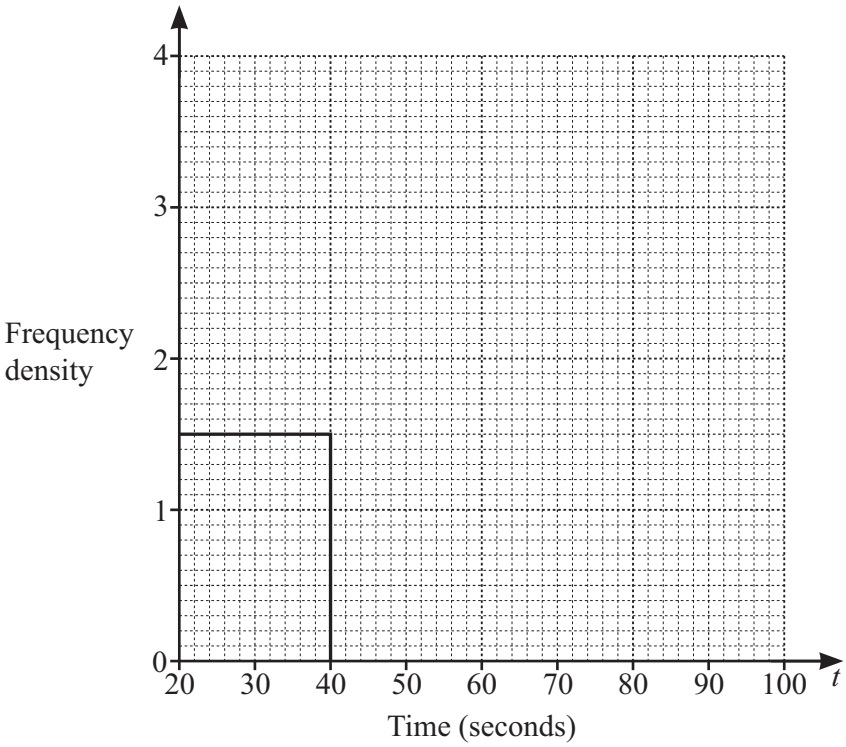
13 **(a)** 100 students solve a puzzle.
The table shows information about the time taken by each student to solve the puzzle.

Time (t seconds)	$20 < t \leqslant 40$	$40 < t \leqslant 60$	$60 < t \leqslant 100$
Frequency	30	40	30

(i) Work out an estimate of the mean.

.....s [4]

(ii) Complete the histogram to show the information in the table.



[2]

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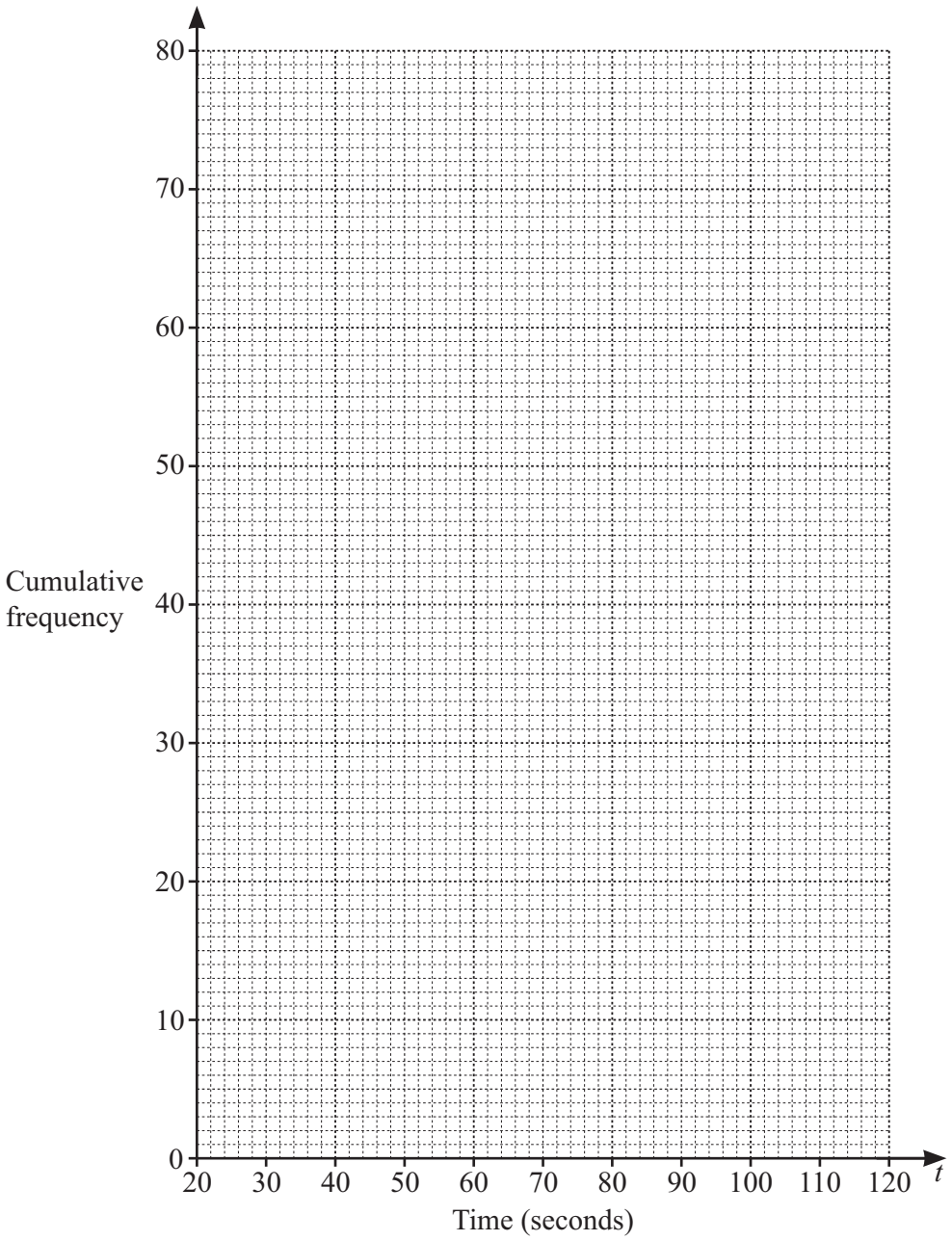
(b) 80 adults solve the same puzzle as the students.
The cumulative frequency table shows information about the time taken by each adult to solve the puzzle.

Time (t seconds)	$t \leqslant 20$	$t \leqslant 40$	$t \leqslant 60$	$t \leqslant 80$	$t \leqslant 100$	$t \leqslant 120$
Cumulative frequency	0	12	36	60	74	80

(i) On the grid, draw a cumulative frequency diagram.



[3]

(ii) Use your cumulative frequency diagram to find an estimate for

(a) the median

..... s [1]

(b) the lower quartile.

..... s [1]

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